OPTICS THESES

Physicist-engineer BSc, 2025

BMEatomfizika

1) Light models, time-dependent vectorial EM wave equation in dielectrics, Helmholtz equation.
2) Light propagation in absorbent media. Complex dielectric permittivity, wave vector and refractive index.
3) Polarization, description by Jones vectors. Uniaxial birefringence, operation of a lambda/4 plate.
4) Description of light propagation in anisotropic media, properties of the dielectric permittivity tensor. Direction of EM field vectors. Connection of phase and ray velocities. Index ellipsoid.
5) Reflection and transmission of a plane wave at a plane interface. Phase matching, Fresnel formulae. Total internal reflection, evanescent wave. Brewster's effect.
6) Geometrical optics. Scope of validity of the model, eikonal equation. Definitions of a light ray, intensity law, Fermat's principle.
7) Paraxial optics, description by matrices. Concept of principal planes, effective focal length, types of magnification. Evaluation of complex systems. Lagrange-Helmholtz equation.
8) Conditions and types of interference. Two-beam interference. Discussion of plane parallel and slightly wedged plates. Definition of visibility.
9) Multiple-beam interference. Resolution of an optical grating. Interferometers (shearing, Fizeau, Michelson, Mach-Zehnder).
10) Statistical optics, temporal coherence. Connection of the normalized coherence function with the interferogram and power spectral density.
11) Quasi-monochromatic waves. Connection between the autocorrelation function and the normalized coherence function. Spatial coherence.
12) Scalar diffraction. Explanation of the Fresnel-Kirchhoff formula and derivation of Fraunhofer and Fresnel approximations.
13) Interpretation of the Fraunhofer approximation. Far-field diffraction pattern of rectangular and circular apertures. Rayleigh criterion, resolution of a telescope.

Gábor Erdei.